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# A Major Project Report
# (Phase-I)

# On

# AI-Based Smart Traffic Congestion Prediction and Signal Optimization System

***Submitted in partial fulfillment of the requirement for the award of the degree of***

## BACHELOR OF TECHNOLOGY

IILM UNIVERSITY

Session 2026-27

in

Computer Science and Engineering

By

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May, 2026

# SCHOOL OF COMPUTER SCIENCE AND ENGG.

## IILM UNIVERSITY, GREATER NOIDA

## CANDIDATE'S DECLARATION

We hereby certify that the work presented in the **Phase-I report** of the project entitled "**AI-Based Smart Traffic Congestion Prediction and Signal Optimization System**", in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in the **School of Computing Science and Engineering, IILM University**, Greater Noida, is an original work carried out by us under the supervision of **Mrs.Rekha baghel**, School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-I report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Signature of Guide Name and Details of Guide

**Name:** MRS.Rekha baghel

**Designation:** _____

**School:** School of Computing Science and Engineering

**IILM University, Greater No**

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# ABSTRACT

Cities continue to experience the problem of traffic congestion further and further, resulting in additional time spent in the car, increased fuel consumption and air pollution. High changes in the old systems with fixed lights and/or simple sensors means that things get backed up quick. The concept is to develop an artificial intelligence system to predict congestion and optimize signals based on actual real-time data.
The combination of machine learning models and computer vision are applied to analyse real-time video streams, counting and categorizing vehicles, and quantifying traffic flow. Past data is mixed with current camera and sensor information to train the models to predict jams a bit in advance. Those forecasts are fed to a control section which adjusts light durations to better manage the load for intersections. The entire set is designed to be portable and so influence a number of spots with control at once or individually. Edge computing helps cut down on delays and less need for strong internet connections. To make the predictions more accurate, things like weather or time of day or events are included. From simulations and some field tests, it seems that better flows are achieved with shorter waiting times and shorter car queues, and lower fuel consumption during idle periods. There may be less emissions as well due to easier movement. When these predictions are combined with the adjustments, a practical approach exists to traffic in urban areas that could reduce the amount of traffic congestion and pollution. I believe the edge part is important here since there are plenty of network problems.

This article discusses the concept of Intelligent Transportation Systems and its role in traffic congestion prediction.The paper explains about Intelligent Transportation Systems and its application in Traffic Congestion Prediction using Artificial Intelligence.
Machine Learning, Deep Learning, Adaptive Traffic Signal Control, Traffic Optimization, Predictive Traffic Management.

# 1. PROBLEM DEFINITION

Traffic demand is continuously changing in urban road networks. At certain times of day, weekends, special events and unexpected conditions, the volume of vehicles approaching an intersection may be significantly different. Conventional
Fixed-time signals have fixed periods, and can lead to unnecessary delay on low traffic approaches, as well as long queues on high traffic approaches. Selected literature confirms the potential of using AI for forecasting and adaptive control, as well as some data scarcity, communication needs, computational cost and scaling problems.

How can current traffic information and the near future traffic condition prediction be structured into a practical and useful AI-based procedure that the predicted traffic condition clearly informs the next traffic signal timing decision, keeping the standard of comparison with a conventional fixed-time signal control?

The proposed system is designed to accomplish four related functions:

(i) analyze current traffic condition
(ii) predict traffic demand or congestion near the future
(iii) use current and predicted traffic information to aid in signal timing or priority decisions, and
(iv) evaluate the system's predicted traffic strategy against fixed-time signal control based on common traffic-performance metrics.

# 2. LITERATURE SURVEY

Urban traffic congestion causes delays, fuel waste, and pollution. Traditional fixed-time signal systems cannot adapt to sudden changes in traffic flow. Early statistical models offered basic predictions but lacked responsiveness.

## 2.1 AI-Powered Traffic Signal Control Systems for Enhanced Traffic Flow and Reduced Waiting Time

Dabi and Abuhamoud (2025) predicts traffic conditions based on historical and real-time traffic data, and adjusts signal timing accordingly. The study created 1000 synthetic samples with time, traffic volume, road condition and weather, the MSE result of the Random Forest experiment was 1.5471. The study is relevant to the proposed project since the use traffic volume, time and available contextual variables are used but the model is validated in realistic scenarios.

## 2.2 Leveraging AI in Urban Traffic Management: Addressing Congestion and Traffic Flow with Intelligent Systems

Nampalli (2021) reviews data-driven traffic monitoring, forecasting and management, and highlights the need for accurate, up-to-date forecasts along with robustness, safety and response-time requirements. The study is pertinent to the proposed project, as it uses historical and real-time traffic data and safety and response time as design constraints.

## 2.3 Implementing AI-Driven Traffic Signal Systems for Enhanced Traffic Management in Dammam

Almatar (2024) Reports an AI-powered traffic management framework for Dammam and finds that adaptive signal control and wireless traffic information can improve allocation of clearing time and traffic-management efficiency. The study is relevant to the proposed project because compare adaptive decisions directly with the fixed-time baseline under controlled scenarios.

## 2.4 Artificial Intelligence based Smart Traffic Enforcement and Management System in urban areas

El-Hansali et al. (2022) Proposes an intelligent dashboard that detects congestion, forecasts using historical data, plans alternative routes and adapts signal timing from current sensor information. The study is relevant to the proposed project because make sensing requirements explicit and leave route planning and enforcement as future extensions.

## 2.5 Artificial Intelligence-Based Adaptive Traffic Signal Control System: A Comprehensive Review

Agrahari et al. (2024) Reviews adaptive traffic signal control at single and multiple intersections and reports substantial potential while identifying continued research needs for more intersections and real-life use. The study is relevant to the proposed project because start with a controlled intersection and design the architecture so it can scale later.

## 2.6 Artificial Intelligence Based Automated Traffic Management System

Sai Pooja, Vaishnavi and Chandra (2021) Proposes camera/IoT-based traffic monitoring

to detect congestion, vehicle density, speed, abnormal traffic patterns and emergency vehicles and use that information for intelligent decisions. The study is relevant to the proposed project because keep the initial implementation modular so emergency and camera-based inputs can be added later.

4

### 2.7 Traffic Management: Implementing AI to Optimize Traffic Flow and Reduce Congestion

Pillai (2024) Explores AI-based predictive traffic management and evaluates models in simulated environments to study congestion reduction and traffic-flow optimization. The study is relevant to the proposed project because use reproducible simulation and compare prediction-driven control against a fixed-time baseline.

### 2.8 Artificial Intelligence for Traffic Management: A Comprehensive Review of Advances and Challenges

Francisco, Robles and Samson (2024) Reviews AI applications for congestion, prediction and management and highlights speed, traffic flow, delay and level of service as useful evaluation dimensions. The study is relevant to the proposed project because use multiple traffic-performance metrics rather than relying on one prediction score.

### 2.9 AI-Driven Traffic Prediction and Congestion Control in Smart Communication Systems

Hussain (2026) Examines AI-based traffic prediction and congestion control in smart communication environments and emphasizes real-time processing, predictive analytics and intelligent communication. The study is relevant to the proposed project because keep the prediction-to-control pipeline computationally practical and document communication assumptions.

### 2.10 Graph neural network for traffic forecasting: A survey

Jiang and Luo (2022) Surveys GNN-based traffic forecasting and emphasizes spatial-temporal dependency modelling, implementation and reproducibility. The study is relevant to the proposed project because use graph-based forecasting as an advanced extension when network-level data are available.

### 2.11 Machine Learning Model Application and Comparison in Actuated Traffic Signal Forecasting

Xie et al. (2023) Compares ML approaches for forecasting actuated traffic signals and shows the value of learning dynamic relationships between traffic flow and signal behaviour. The study is relevant to the proposed project because compare more than one ml predictor before final model selection.

### 2.12 Adaptive urban traffic signal control based on enhanced deep reinforcement learning

Cai and Wei (2024) Uses dueling networks, double Q-learning, prioritized sampling and noise to improve convergence and robustness for adaptive signal control. The study is relevant to the proposed project because use drl as an advanced controller only after the baseline prediction and fixed-time experiments are stable.

### 2.13 LSTTN: A Long-Short Term Transformer-based spatiotemporal neural network for traffic flow forecasting

Luo et al. (2024) Models long- and short-term traffic patterns and addresses the difficulty of learning useful representations from long traffic histories. The study is relevant to the proposed project because provides an advanced forecasting benchmark when sufficient data and computation are available.

### 2.14 XLight: An interpretable multi-agent reinforcement learning approach for traffic signal control

Cai, Fang and Xu (2025) Introduces interpretable state, action and reward design for network-wide TSC and evaluates the approach on isolated, synthetic and real-world network scenarios. The study is relevant to the proposed project because make the proposed decision layer transparent and keep learned control decisions auditable.

### 2.15 Reinforcement learning for traffic signal control in large-scale transportation networks: a systematic literature review

Zhang et al. (2026) reviewed RL-based traffic signal control and highlighted the importance of controlled simulations before scaling to complex, multi-intersection networks.

# 2. LITERATURE SURVEY

### 2.16 Overall Literature Findings

As can be seen in the selected studies, the evolution is from traditional fixed-time signal control to data-driven and adaptive traffic management. Dabi and Abuhamoud correlate the predicted duration of the signals directly with machine-learning prediction and consider the traffic volume, time, road condition, and weather. Nampalli stresses the importance of precise and up-to-date predictions and strong architectures for intelligent transportation. Almatar and El-Hansali et al. reiterate the need for sensors, wireless communication and adaptive control for urban traffic networks. Agrahari et al. combine fuzzy logic, metaheuristics, dynamic programming, reinforcement learning and deep reinforcement learning and point out scalability and multi-intersection coordination as open problems. At the same time, Sai Pooja et al. focus on camera/IoT-based monitoring and emergency-vehicle identification, and Pillai and Francisco et al. on AI, neural networks and computer vision for predictive traffic management. Hussain takes the conversation further to intelligent communication environments such as IoT, edge/fog computing and real-time constraints.

### 2.17 Advanced Prediction and Control Directions

Jiang and Luo demonstrate how GNNs can be applied to model the structure of spatial-temporal traffic. Xie et al. argue that machine-learning techniques for actuated signal forecasting can be directly compared. Cai and Wei show that deep reinforcement learning continues to have a significant role in adaptive signal control, while Luo et al.

To demonstrate the importance of spatiotemporal forecasting using transformers. Cai, Fang and Xu contribute an important practical perspective by focusing on interpretability in multi-agent traffic-signal control. Zhang et al. provides a review of reinforcement learning in large-scale traffic networks and underscores the need for dealing with scalability and the gap between simulation and real-world.

### 2.18 Research Gap for the Proposed Project

An explicit prediction-to-signal pipeline is still a useful contribution to the project level: prediction should influence the next signal decision, not be explicitly reported as a stand-alone output.

The choice of a model must be based on data quality and data availability. The project should only contain variables contained in the chosen data set and explicit limitations should be stated.

There is coordination and communication complexity added in multi-intersection and multi-agent methods. The first implementation should thus be targeted in one intersection or in a small network which can be controlled.

Operational traffic measures (average delay, waiting time, queue length, throughput, congestion) as well as prediction measures (MAE and RMSE) should be used to evaluate performance.

The results of simulations should be given as evidence of controlled experiments, rather than of effectiveness in the real city scale.

## Table 2.1 Comparative Summary of Reviewed Research Papers

| Work | Year | Data / Scope | Main AI / Method | Key Result | Major Challenge | Lesson |
|---|---|---|---|---|---|---|
| Dabi et al. | 2025 | 1000 synthetic traffic samples + real-time input | Random Forest / machine learning with real-time traffic data | Uses historical and real-time traffic information to predict future traffic conditions and dynamically adjust signal timing. The study generated 1000 synthetic samples using time, traffic volume, road condition and weather; its Random Forest experiment reported MSE 1.5471. | Synthetic traffic data and omission of special events and severe real-world conditions. | Use traffic volume, time and available contextual variables, but validate the model in realistic scenarios. |
| Nampalli | 2021 | Historical + real-time / ITS literature | AI / ML / deep learning / ITS review | Reviews data-driven traffic monitoring, forecasting and management and emphasizes accurate, up-to-date forecasts together with robustness, safety and response-time requirements. | Data quality, safety, response time and communication between networks. | Combine historical and real-time traffic information and treat safety and response time as design constraints. |
| Almatar | 2024 | Dammam traffic modelling + wireless data | Fuzzy inference, traffic modelling, wireless communication and adaptive signal control | Reports an AI-powered traffic management framework for Dammam and finds that adaptive signal control and wireless traffic information can improve allocation of clearing time and traffic-management efficiency. | Data quality, infrastructure, model assumptions, cost and simulation-to-real-world gap. | Compare adaptive decisions directly with the fixed-time baseline under controlled scenarios. |
| El-Hansali et al. | 2022 | Historical data + sensors / RFID | AI, machine learning, RFID, sensors and predictive management | Proposes an intelligent dashboard that detects congestion, forecasts using historical data, plans alternative routes and adapts signal timing from current sensor information. | Large-scale sensing, communication and infrastructure requirements. | Make sensing requirements explicit and leave route planning and enforcement as future extensions. |
| Agrahari et al. | 2024 | ATSC literature; single and multiple intersections | Fuzzy logic, metaheuristics, dynamic programming, RL, DRL and hybrids | Reviews adaptive traffic signal control at single and multiple intersections and reports substantial potential while identifying continued research needs for more intersections and real-life use. | Scalability, multi-intersection coordination and real-world application. | Start with a controlled intersection and design the architecture so it can scale later. |

Table 2.1

| | | | | | | |
|---|---|---|---|---|---|---|
| Sai Pooja, Vaishnavi and Chandra | 2021 | Cameras + IoT traffic information | AI, ML, IoT and computer vision | Proposes camera/IoT-based traffic monitoring to detect congestion, vehicle density, speed, abnormal traffic patterns and emergency vehicles and use that information for intelligent decisions. | Data availability, communication and unusual traffic events. | Keep the initial implementation modular so emergency and camera-based inputs can be added later. |
| Pillai | 2024 | Traffic data + simulated environments | Machine learning, neural networks and computer vision | Explores AI-based predictive traffic management and evaluates models in simulated environments to study congestion reduction and traffic-flow optimization. | Scalability and dependence on simulated evaluation. | Use reproducible simulation and compare prediction-driven control against a fixed-time baseline. |
| Francisco et al. | 2024 | AI traffic-management literature | AI / ML / deep learning / CNN / RNN / LSTM | Reviews AI applications for congestion, prediction and management and highlights speed, traffic flow, delay and level of service as useful evaluation dimensions. | Prediction accuracy, scalability, data quality and environmental impact. | Use multiple traffic-performance metrics rather than relying on one prediction score. |
| Hussain | 2026 | IoT / 5G-6G / edge traffic environments | ML, deep learning, RL, IoT, 5G/6G, edge/fog computing | Examines AI-based traffic prediction and congestion control in smart communication environments and emphasizes real-time processing, predictive analytics and intelligent communication. | Data heterogeneity, computational complexity, scalability, privacy and real-time limitations. | Keep the prediction-to-control pipeline computationally practical and document communication assumptions. |
| Jiang et al. | 2022 | Traffic forecasting studies / public datasets | Graph neural networks | Surveys GNN-based traffic forecasting and emphasizes spatial-temporal dependency modelling, implementation and reproducibility. | Model complexity and cross-dataset generalization. | Use graph-based forecasting as an advanced extension when network-level data are available. |
| Xie et al. | 2023 | Actuated traffic signal forecasting data | Machine-learning model comparison | Compares ML approaches for forecasting actuated traffic signals and shows the value of learning dynamic relationships between traffic flow and signal behaviour. | Dependence on traffic data quality and transferability to different signal systems. | Compare more than one ML predictor before final model selection. |
| Cai et al. | 2024 | Traffic-signal simulation | Enhanced deep reinforcement learning | Uses dueling networks, double Q-learning, prioritized sampling and noise to improve convergence and robustness for adaptive | Training efficiency and robustness across traffic conditions. | Use DRL as an advanced controller only after the baseline prediction and fixed-time experiments are stable. |

Table 2.1

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | signal control. | | |
| Luo et al. | 2024 | METR-LA, PEMS-BAY, PEMS04, PEMS08 | Transformer + spatiotemporal neural network | Models long- and short-term traffic patterns and addresses the difficulty of learning useful representations from long traffic histories. | Long-range dependency modelling and computational cost. | Provides an advanced forecasting benchmark when sufficient data and computation are available. |
| Cai, Fang and et al. | 2025 | Synthetic + real-world network TSC scenarios | Interpretable multi-agent reinforcement learning | Introduces interpretable state, action and reward design for network-wide TSC and evaluates the approach on isolated, synthetic and real-world network scenarios. | Interpretability and practical deployment of learned controllers. | Make the proposed decision layer transparent and keep learned control decisions auditable. |
| Zhang et al. | 2026 | Large-scale TSC literature | Systematic review of RL for large-scale traffic signal control | Reviews RL-based TSC for large networks and synthesizes simulation environments, transportation modalities and methodological advances. | Large-scale coordination and simulation-to-real-world transfer. | Keep Phase-I controlled and reproducible before moving toward multi-intersection city-scale experiments. |

Table 2.1

# 4. PROPOSED CONTRIBUTION AND NOVELTY

## 1. Proposed Contribution

Design a modular AI-driven traffic management system that includes the existing traffic analysis, traffic congestion prediction, signal optimization and signal performance evaluation.

Instead of predicting as an isolated output, use the predicted near-future congestion as an explicit input to the traffic-signal decision.

Implement a clear signal-priority/signal-time allocation system based on up-to-date traffic data and forecasted demand.

Compare prediction-driven dynamic control to conventional fixed-time control, under same traffic conditions. Test the system with average delay, waiting time, length of the queue, throughput and congestion.

## 2. Novelty of the Proposed Work

25

AI-based traffic prediction, adaptive signal control, DRL, and IoT-assisted monitoring and network-wide coordination have already been well established in the literature. Not that this is a novel idea; rather, its novelty lies in the way this project is being implemented. Rather, the suggested contribution is a clear and predictable decision pipeline in which near future congestion is used as a direct and measurable variable to determine signal timing and then a fair comparison is made with the fixed-time control under identical conditions.

## 3. Proposed System Workflow

1. **Traffic Data Input**
   Vehicle count, density, speed, queue length, time and other available traffic features.
2. **Current Traffic Analysis**
   Estimate the present traffic condition at the selected intersection or road approaches.
3. **Near-Future Prediction**
   Forecast future traffic demand or congestion using an appropriate AI/ML model.
4. **Signal Optimization**
   Use current and predicted traffic conditions to select signal priority or timing.
5. **Performance Evaluation**
   Compare dynamic prediction-driven control with fixed-time control using common traffic metrics.

# 5. OBJECTIVES

1.To study recent AI-based approaches for traffic congestion prediction and adaptive traffic signal control.

2.To identify research-paper-supported limitations and practical research gaps relevant to the proposed project.

3.To collect and preprocess suitable traffic data for current traffic analysis and near-future congestion prediction.

4.To develop and evaluate an AI/ML model for predicting future traffic demand or congestion.

5.To design a modular decision layer in which predicted congestion explicitly influences traffic-signal timing.

6.To implement a prediction-driven dynamic signal-control strategy in a controlled experimental environment.

7.To compare the proposed strategy with conventional fixed-time signal control under identical traffic scenarios.

8.To evaluate performance using average delay, waiting time, queue length, throughput and congestion level, while also reporting suitable prediction metrics such as MAE and RMSE.

# 6. SCOPE

1. The initial project focuses on AI-based urban intersection traffic management.
2. The implementation may use suitable public traffic datasets, simulation-generated data, or a combination of both.
3. The first implementation will use traffic variables supported by the selected dataset rather than assuming the availability of every possible sensor modality.
4. The prediction horizon will be explicitly defined during model development.
5. The signal-control component will use current and predicted traffic information through a transparent decision mechanism.
6. Conventional fixed-time signal control will be used as the primary baseline for comparison.
7. The initial experimental setup may focus on a single intersection or a controlled small-scale scenario.
8. The modular architecture can later be extended with real-time camera feeds, connected-vehicle data, weather information, pedestrian movement, emergency-vehicle priority and multi-agent coordination.
9. Full city-scale field deployment is outside the Phase-I scope unless an appropriate real-world pilot environment becomes available.

# 7. OBJECTIVES

1.To study recent AI-based approaches for traffic congestion prediction and adaptive traffic signal control.

2.To identify research-paper-supported limitations and practical research gaps relevant to the proposed project.

3.To collect and preprocess suitable traffic data for current traffic analysis and near-future congestion prediction.

4.To develop and evaluate an AI/ML model for predicting future traffic demand or congestion.

5.To design a modular decision layer in which predicted congestion explicitly influences traffic-signal timing.

6.To implement a prediction-driven dynamic signal-control strategy in a controlled experimental environment.

7.To compare the proposed strategy with conventional fixed-time signal control under identical traffic scenarios.

8.To evaluate performance using average delay, waiting time, queue length, throughput and congestion level, while also reporting suitable prediction metrics such as MAE and RMSE.

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